

Weekly Report: [W05]

General Information

* **Group ID:** 52
* **Group Name:** Group 52
* **Class:** 25A01
* **Project Name:** Super Mario Game
* **Date range:** 2026-07-05 – 2026-07-11
* **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

Branch: `A/physics-input-test` (Physics Refactoring & Camera System)

* **Task 1: Full Camera Class Implementation:** Coded the implementation of the `Camera` class in `Camera.cpp`. Integrated LERP smooth following to follow character coordinates, random coordinate offsets for screen shake effects, and viewport clamping boundaries that center the screen when level sections are smaller than viewport dimensions (`1280x720`).

* **Task 2: Decoupled Physics Command Intents:** Modified horizontal movement and sprint commands (`MoveLeftCommand`, `MoveRightCommand`, `RunCommand`) to register intents as boolean requests on characters/players rather than applying direct velocity modifications. This maintains clean command pattern separation.

* **Task 3: Momentum and Friction Centralization:** Moved passive horizontal deceleration/friction, active horizontal acceleration, and run speed limit scaling directly into the `PhysicsEngine::update()` loop. Surface properties (like low friction on Ice tiles) are read and calculated dynamically.

* **Task 4: Camera & Playground Integration:** Declared and initialized `Camera` inside `PlayingState`. Updated `PlayingState::render()` to run dual-view rendering, setting the active camera view to draw world assets in scrollable space and default view for screen overlays.

* **Task 5: Mid-Air Wall Friction & Sliding Mechanics:** Added slide friction inside `resolveEntityVsTile()`. When a character hits a wall in mid-air, any upward momentum (`velocity.y < 0`) is zeroed out to prevent the player from gliding up walls. Falling speed is capped at `Constants::WALL\_SLIDE\_SPEED = 50.0f` px/s.

25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems)

Branch: `B/feature/enemies` (Gameplay Entities - In Progress)

* **Task 1: AI Movement Strategy Patterns:** Integrating movement behaviors (patrolling, proximity chases) for basic enemies.

* **Task 2: Concrete Enemy & Block Implementations:** Writing concrete class functions for base Goombas and interactive block types.

AI Usage Declaration

* Used AI to troubleshoot coordinate scaling issues between scrollable camera views and static screen space rendering.

* Used AI to design the internal edge collision algorithm that filters horizontal floor boundary crossings.

* Used AI to refine the timing sequence of `onGround` status resets inside the physics pipeline.

Issues & Resolutions

Issue 1: Internal Edge Collision Bug (Stuck on Flat Ground)

* **Problem**: When walking across flat floors composed of adjacent tiles, the player would randomly stop or stick at tile seams. This occurred because horizontal overlaps at the border were misidentified as horizontal side wall collisions (`normal.x != 0.0f`), which cancelled `velocity.x` and set `onWall = true`.

* **Resolution**: Added internal edge checking inside `CollisionDetector::checkEntityVsTileMap()`. If a horizontal collision is detected on a tile, we check the tile immediately above it. If that upper tile is empty/non-solid, we know this is a flat floor seam, not a wall, and dynamically convert it into a vertical push-out, preserving horizontal velocity.

**Issue 2: Ground Flag Jittering & Stale Bounding Box Sync

* **Problem**: Standing or walking on tiles caused the `onGround` state to rapidly toggle between `TRUE` and `FALSE` between consecutive frames, causing the player to jitter slightly.

* **Resolution**: Found that when `resolveEntityVsTile` updated `entity.position`, it did not synchronize the `boundingBox` coordinates. Stale bounding boxes remained inside the tile on the next frame, re-triggering overlaps. Added immediate `boundingBox` synchronization steps upon any resolution displacement.

**Issue 3: Double-Resolution Cumulative Push-outs

* **Problem**: When a player spanned two adjacent floor tiles, both floor tile collisions were resolved sequentially. The push-outs were summed together (e.g. pushed up twice), throwing the player slightly into the air, which immediately set `onGround = false`.

* **Resolution**: Refactored the Y and X collision resolution steps inside `PhysicsEngine::update()` to find and resolve only the single collision containing the **maximum overlap** (largest penetration depth) along each axis per frame. Resolving only the maximum overlap is mathematically sufficient to clear all overlaps on flat ground without double-counting pushes.

**Issue 4: Wall Momentum Retention (Air Wall Collision)

* **Problem**: Jumping and holding movement keys against a vertical wall allowed the player to glide upwards along the wall with full momentum.

* **Resolution**: Added wall friction inside `resolveEntityVsTile()`. If a player collides horizontally with a wall in mid-air (`!onGround`), their upward vertical velocity (`velocity.y < 0.0f`) is zeroed out to simulate friction/momentum loss. Falling velocity is also capped at `WALL_SLIDE_SPEED = 50.0f` px/s.